

BT-102 – MATHEMATICS-I

UNIT-WISE 5 MOST IMPORTANT TOPICS

(Analysis based on 2019–2025 papers)

MODULE-1: Differential Calculus

1. **Rolle's Theorem / Lagrange's Mean Value Theorem (Statement + Verification)**
→ May-2019, Jun-2020, Nov-2022, Dec-2024
 2. **Taylor / Maclaurin Series Expansion (one variable)**
→ Nov-2019, Jun-2020, Dec-2023, Jun-2025
 3. **Partial Differentiation (higher order derivatives)**
→ May-2019, Dec-2020, Jun-2023
 4. **Maxima & Minima of functions of two variables**
→ Nov-2019, Dec-2023, Jun-2025
 5. **Lagrange Multipliers (constrained optimization)**
→ Dec-2020, Nov-2022, Dec-2024
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MODULE-2: Integral Calculus

1. **Definite Integral as Limit of Sum / Properties**
→ May-2019, Jun-2020, Dec-2023
2. **Beta & Gamma Functions (evaluation of integrals)**
→ Nov-2019, Nov-2022, Dec-2024
3. **Change of Order of Integration (double integrals)**
→ Jun-2020, Jun-2023, Jun-2025
4. **Applications of Definite Integrals (Area / Volume)**

→ Dec-2020, Nov-2022, Dec-2024

5. Multiple Integrals (area/volume calculation)

→ May-2019, Dec-2023

MODULE-3: Sequences & Series

1. Tests of Convergence (Comparison / D'Alembert / Cauchy)

→ Nov-2019, Jun-2020, Nov-2022

2. Power Series & Taylor Series

→ Dec-2020, Jun-2023, Dec-2024

3. Expansion of functions (exp, log, trig)

→ May-2019, Nov-2022

4. Fourier Series (full range)

→ Dec-2023, Jun-2025

5. Half-range Fourier Series / Parseval's Theorem

→ Dec-2020, Dec-2024

MODULE-4: Vector Spaces

1. Vector Space & Subspace (proof-based)

→ May-2019, Jun-2020, Nov-2022

2. Linear Dependence & Independence

→ Dec-2020, Jun-2023

3. Basis & Dimension of Vector Space

→ Nov-2019, Dec-2024

4. Linear Transformations (verification)

→ Jun-2020, Dec-2023

5. Rank-Nullity Theorem (application)

→ Nov-2022, Jun-2025

MODULE-5: Matrices & Linear Algebra

1. Rank of Matrix (elementary transformations)

→ May-2019, Dec-2020, Jun-2023

2. Consistency of Linear Equations

→ Nov-2019, Dec-2023, Dec-2024

3. Eigen Values & Eigen Vectors

→ Jun-2020, Nov-2022, Jun-2025

4. Diagonalization of Matrices

→ Dec-2020, Jun-2023

5. Cayley–Hamilton Theorem (finding inverse/powers)

→ Nov-2019, Dec-2024, Jun-2025

FULL PREDICTED MODEL QUESTION PAPER

Time: 3 Hours

Maximum Marks: 70

Instructions:

1. Attempt **any FIVE** questions.
 2. All questions carry **equal marks (14 marks each)**.
 3. Use standard notations and show all necessary steps.
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Q1. (MODULE-1: Differential Calculus)

a) State and verify **Lagrange's Mean Value Theorem** for

$f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

(7)

b) Expand $\log(1 + x)$ using **Taylor/Maclaurin series** up to four terms.

(7)

Q2. (MODULE-1: Partial Differentiation & Optimization)

a) If

$$u = x^2 \tan^{-1}\left(\frac{y}{x}\right) + y^2 \tan^{-1}\left(\frac{x}{y}\right),$$

find

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}.$$

(7)

b) Find the maximum and minimum values of

$$f(x, y) = x^2 + y^2 + 2x + 4y + 5.$$

(7)

Q3. (MODULE-2: Integral Calculus)

a) Evaluate using **Beta and Gamma functions**:

$$\int_0^1 x^3(1-x)^2 dx.$$

(7)

b) Change the order of integration and evaluate:

$$\int_0^1 \int_0^{\sqrt{1-x^2}} xy \, dy \, dx.$$

(7)

Q4. (MODULE-3: Sequences & Fourier Series)

a) Test the **convergence** of the series:

$$\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}.$$

(7)

b) Find the **Fourier series** of

$f(x) = x$ in the interval $(-\pi, \pi)$.

(7)

Q5. (MODULE-4: Vector Spaces)

a) Show that the set

$$W = \{(x, y, z) : x + 2y + 3z = 0\}$$

is a **subspace** of $\mathbb{R}^3$.

(7)

b) Determine whether the vectors

$$(1, 2, 3), (2, 4, 6), (1, 1, 1)$$

are **linearly independent** or **dependent**.

(7)

Q6. (MODULE-5: Matrices – Rank & Consistency)

a) Find the **rank** of the matrix using elementary transformations:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix}.$$

(7)

b) Test the **consistency** and solve the system of equations:

$$\begin{aligned} x + 2y + z &= 4 \\ 2x + 5y + 3z &= 9 \\ x + y + 2z &= 5 \end{aligned}$$

(7)

Q7. (MODULE-5: Eigen Values & Cayley–Hamilton)

a) Find the eigen values and eigen vectors of

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

(7)

b) Verify Cayley–Hamilton theorem for

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

and hence find A^{-1} .

(7)

Q8. (MODULE-2 / 3: Applications & Series)

a) Find the area enclosed between the curves

$$y = x^2 \text{ and } y = 2x.$$

(7)

b) Obtain the half-range sine series for

$$f(x) = x \text{ in } (0, \pi).$$

(7)

PREDICTED MODEL QUESTION PAPER – SET-2

Time: 3 Hours

Maximum Marks: 70

Instructions:

1. Attempt **any FIVE** questions.
 2. All questions carry **equal marks (14 each)**.
 3. Show necessary steps for full credit.
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Q1. (MODULE-1: Mean Value Theorem & Series)

a) Verify Rolle's Theorem for

$$f(x) = x^3 - 3x + 2 \text{ on the interval } [-2, 1].$$

(7)

b) Expand e^x using Maclaurin series up to four non-zero terms.

(7)

Q2. (MODULE-1: Partial Differentiation & Lagrange Multipliers)

a) If

$$u = x^2y + y^2x,$$

find

$$\frac{\partial^2 u}{\partial x \partial y}.$$

(7)

b) Find the **extreme values** of

$$f(x, y) = xy$$

subject to the constraint

$$x^2 + y^2 = 1$$

using **Lagrange's Multipliers**.

(7)

Q3. (MODULE-2: Definite Integrals & Beta–Gamma)

a) Evaluate the definite integral:

$$\int_0^{\pi/2} \sin^3 x \cos^2 x \, dx.$$

(7)

b) Evaluate using **Gamma function**:

$$\int_0^{\infty} x^4 e^{-x} \, dx.$$

(7)

Q4. (MODULE-2: Multiple Integrals)

a) Change the order of integration and evaluate:

$$\int_0^1 \int_x^1 (x + y) dy dx.$$

(7)

b) Find the area of the region bounded by

$$y^2 = 4x \text{ and } x = 4.$$

(7)

Q5. (MODULE-3: Sequences & Fourier Series)

a) Test the convergence of the series:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

(7)

b) Find the half-range cosine series for

$$f(x) = x \text{ in } (0, \pi).$$

(7)

Q6. (MODULE-4: Vector Spaces & Linear Transformation)

a) Determine whether the set

$$V = \{(x, y, z) : x = y = z\}$$

is a **subspace** of $\mathbb{R}^3$.

(7)

b) Show that the transformation

$$T(x, y, z) = (x + y, y + z)$$

is **linear** and find its **kernel**.

(7)

Q7. (MODULE-5: Rank & Eigen Values)

a) Find the **rank** of the matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 1 & 1 & 2 \end{bmatrix}$$

using elementary transformations.

(7)

b) Find the **eigen values** of

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

(7)

Q8. (MODULE-5: Cayley–Hamilton Theorem)

a) Verify Cayley–Hamilton theorem for

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

(7)

b) Using Cayley–Hamilton theorem, find A^{-1} .

(7)
